

Infron vs OpenRouter Routing, Cache, Cost, and Streaming Performance A/B Report

Executive Summary

This report evaluates `openai/gpt-5.4-mini` across Infron and OpenRouter in prompt-caching workloads. The experiment uses only `/v1/chat/completions`, keeps model/platform default reasoning and thinking behavior, and keeps prompt-length stratification enabled.

After A/B filtering, the analysis retains **792 paired samples** and **3168 request-level observations**. It excludes **16 records**.

Core finding: Infron leads throughput in every routing mode and has lower cost in most routing modes. OpenRouter generally has lower E2E latency and Streaming TTFT. Cache hit rates are high on both platforms, with route-specific winners.

Quality Gate

Item	Value
Model	<code>openai/gpt-5.4-mini</code>
API protocol	<code>/v1/chat/completions</code> only
Groups / rounds	4 groups x 50 rounds
Prompt tiers	<code>short</code> about 1500, <code>medium</code> about 8000, <code>long</code> about 32000 tokens
Reasoning / thinking	Default platform/model behavior; no explicit disable parameter
Pairing rule	strict sort/group/round pair with first/second usage.prompt_tokens deltas <= 50
Retained pairs	792
Request-level rows	3168
Excluded records	16

Route–Level Result Matrix

Routing mode	Pairs	Token cache hit rate	Observed cost	Throughput	E2E latency	TTFT
Throughput First	200	Infron 96.68% / OpenRouter 97.91%	Infron \$0.665196 / OpenRouter \$0.736837	Infron 17.822 tok/s / OpenRouter 5.076 tok/s	Infron 4042.33 ms / OpenRouter 2737.70 ms	Infron 2722.39 ms / OpenRouter 2337.97 ms
Price First	200	Infron 97.99% / OpenRouter 97.95%	Infron \$0.549838 / OpenRouter \$0.643934	Infron 18.259 tok/s / OpenRouter 4.922 tok/s	Infron 3896.09 ms / OpenRouter 2830.58 ms	Infron 2763.05 ms / OpenRouter 2492.73 ms
Latency First	192	Infron 97.98% / OpenRouter 96.73%	Infron \$0.514386 / OpenRouter \$0.642328	Infron 20.873 tok/s / OpenRouter 5.189 tok/s	Infron 3465.18 ms / OpenRouter 2690.54 ms	Infron 2473.68 ms / OpenRouter 2350.08 ms
TTFT First	200	Infron 96.43% / OpenRouter 97.99%	Infron \$0.868611 / OpenRouter \$0.579755	Infron 25.861 tok/s / OpenRouter 6.077 tok/s	Infron 2694.75 ms / OpenRouter 2301.99 ms	Infron 2141.42 ms / OpenRouter 2098.17 ms

API Protocol Record

API protocol	Endpoint	Note
chat_completions	/v1/chat/completions	This run keeps only the standard Chat Completions protocol; it does not include /v1/messages or /v1/responses .

Prompt–Length Stratified Cache Performance

Tier	Pairs	Token cache hit rate	Call cache hit rate	E2E latency	TTFT	Reasoning tokens
short		Infron 0.00% / OpenRouter 0.00%	Infron 0.00% / OpenRouter 0.00%	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0 / OpenRouter 0
medium		Infron 0.00% / OpenRouter 0.00%	Infron 0.00% / OpenRouter 0.00%	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0 / OpenRouter 0
long		Infron 0.00% / OpenRouter 0.00%	Infron 0.00% / OpenRouter 0.00%	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0 / OpenRouter 0

Reasoning / Thinking Observation

This run does not explicitly disable or force reasoning/thinking. Reasoning tokens are treated as observed telemetry.

Routing mode	Reasoning tokens	Avg reasoning tokens/request
Throughput First	Infron 0 / OpenRouter 0	Infron 0.0000 / OpenRouter 0.0000
Price First	Infron 0 / OpenRouter 0	Infron 0.0000 / OpenRouter 0.0000
Latency First	Infron 0 / OpenRouter 0	Infron 0.0000 / OpenRouter 0.0000
TTFT First	Infron 0 / OpenRouter 0	Infron 0.0000 / OpenRouter 0.0000

Reproducibility References

Artifact	GitHub path	SHA256 / Notes
Summary	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/summary.json	b5fcdd6787abbc331d2a4326ce6473a0bd40226faa9adc3bf370bfa1c3
Paired dataset CSV	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/benchmark_pairs.csv	3bd26737e7ca08438856d3d9f6367df63ac391adc7e0db841c81ae34a3
Request-level dataset JSONL	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/benchmark_requests.jsonl	4ff92a2e44c3d37cf6eb732ca00a85393114c4712ec35780b884899c55
Excluded-record audit	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/records_excluded.json	11e2ed24815a7cf38765b2149b9d28bd51f312a36dbcedad6ef6d32b16

Artifact	GitHub path	SHA256 / Notes
Benchmark runner source	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/rerun_routing_sort_cache_cost_ab.py	a485c5116e99375e6b171d0328e5a423babd1ee6b206c72f54053a3a04
HTML report renderer source	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/render_glm52_deepseek_style_report.py	e5bb0d32361d70ee688e0d0a0da3302a1e542e33e53876c905814eb5c2
Test source	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/test_rerun_routing_sort_cache_cost_ab.py	c086ddc5d0a9a91eba82b7e8767d7bddf2f1ca4a28af0a87056b702882
Reports directory	https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports	Bilingual HTML / Markdown / PDF reports
Data directory	https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data	Pair data, request telemetry, summary, and excluded records
Manifest	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/metadata/manifest.json	File sizes and SHA-256 checksums

Dataset reference: `business_representative` built-in representative business templates; request-level export is `benchmark_requests.jsonl`.

Online HTML: Chinese https://infronai.github.io/prompt-cache-bench/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports/routing-cache-cost-streaming-performance-ab-study_gpt-5-4-mini_infron-vs-openrouter_4x50-stream-ttft-reasoning-default-length-tiers-chat-completions_2026-07-04.zh.html; English https://infronai.github.io/prompt-cache-bench/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports/routing-cache-cost-streaming-performance-ab-study_gpt-5-4-mini_infron-vs-openrouter_4x50-stream-ttft-reasoning-default-length-tiers-chat-completions_2026-07-04.en.html.